

Topic: Introduction to Relational Algebra

Q. What is Relational Algebra in DBMS Explain its basic concepts and categorize the primary relational algebra operators.

[DEF] Definition to Remember

Relational Algebra is a procedural query language that forms the theoretical foundation of SQL. It specifies both *what* data is needed and *how* to retrieve it through a sequence of operations on relations (tables), where every operation takes one or two relations as input and produces a new relation as output.

Introduction to Relational Algebra

Concept	Explanation
Input/Output	Every operation takes 1 or 2 relations
Closure Property	Since output is always a relation, operations are closed
No Side Effects	Operations only query data -- they do not modify it

2. Relational Algebra Operators

A. Basic (Fundamental) Operators

Operator	Symbol	Description
Selection	σ	Extracts specific rows satisfying a condition
Projection	π	Extracts specific columns , discards others
Union	$\cup$	Combines rows from two relations, removes duplicates
Set Difference	$-$	Rows in first relation NOT in the second
Cartesian Product	$\times$	Combines every row of A with every row of B
Rename	ρ	Renames a relation or its attributes

B. Derived Operators

Operator	Symbol	Description
Intersection	$\cap$	Rows present in both relations (disjoint sets)
Join	JOIN	Combines related tuples from two relations
Division	$\div$	Answers "for all" queries (e.g., student who has scored more than all others in a subject)

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Relational Algebra is a procedural query language -- the mathematical foundation of SQL.
2. Every operation takes 1 or 2 relations as input and always outputs a new relation (Closure Property).
3. Operations do not modify the original tables -- they are read-only.
4. Basic operators: Selection (σ), Projection (π), Union ($\cup$), Set Difference ($-$), Cartesian Product ($\times$), Rename (ρ).
5. Derived operators: Intersection ($\cap$), Join (JOIN), Division ($\div$).
6. Derived operators can be expressed using combinations of basic operators.
7. Understanding Relational Algebra helps understand how the DBMS engine executes and optimizes SQL queries.

[RECALL] Quick Recall

Procedural Language -> Closure Property -> Basic (σ , π , U , $\bowtie$, $\Join$) -> Derived (ρ , $\Join$, $\Join$) -> Foundation of SQL